

● HARD

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

01

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

$$xkx - 56 = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

One solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. What is the value of k ?

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

NONLINEAR EQUATIONS IN ONE VARIABLE AND SYSTEMS OF EQUATIONS IN TWO VARIABLES — SET 1 · ADVANCED MATH 39

Q3

ADVANCED MATH

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation, c is a positive constant. Which of the following is one of the solutions to the given equation?

- A. $-c$
- B. $-c^2 - 39^2$
- C. $-\sqrt{39^2 - c^2}$
- D. $-\sqrt{c^2 + 39^2}$

Q4

ADVANCED MATH

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

- A. $p + r + s$
- B. $20p + r + s$
- C. $\frac{prs}{pr + ps + rs}$
- D. $\frac{prs}{20p + 20r + 20s}$

Q5

GRID-IN

ADVANCED MATH

$$xx + 1 - 56 = 4xx - 7$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN

ADVANCED MATH

In the xy -plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a) . What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

ADVANCED MATH

$$\frac{14x}{7y} = 2\sqrt{w+19}$$

The given equation relates the distinct positive real numbers w , x , and y . Which equation correctly expresses w in terms of x and y ?

- A. $w = \sqrt{\frac{x}{y}} - 19$
- B. $w = \sqrt{\frac{28x}{14y}} - 19$
- C. $w = \frac{x^2}{y} - 19$
- D. $w = \frac{28x^2}{14y} - 19$

Q8

ADVANCED MATH

$$-9x^2 + 30x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. 3
- B. 0
- C. -25
- D. -53

Q9

GRID-IN

ADVANCED MATH

$$-x^2 + bx - 676 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab , where k is a constant. What is the value of k ?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57